PYTHON PROGRAMMING



21 Days Challenge

Day-11

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FUNCTION FOUNDATIONS - CALLING FUNCTIONS, FUNCTION PARAMETERS & ARGUMENTS, RETURN VALUES



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Function Definition

A block of code that runs when called

- Functions help reuse code without rewriting it.
- Functions can take inputs (parameters) or return outputs.



Creating & Calling a Function

```
def say_hello():
    print("Hello!")

say_hello()
```

Arguments (Parameters)

Parameter = variable in function definition

Argument = value we give when calling function

```
def greet(name):
    print("Hello " + name)

greet("Abdullah")
```

Multiple Arguments

```
def about(name,age):
    print(f"I am {name} and I am {age} years old")

about("abdullah", 28)
```

Output:

```
I am abdullah and I am 28 years old
```

Arbitrary Arguments

Use *args when you don’t know how many arguments will be passed

```
def about(*info):
    for i in info:
        print(i)
about("zubaer", 28, 3.60, "Dhaka")
```

Output:

```
Zubaer
28
3.6
Dhaka
```

Keyword Arguments

Keyword Arguments are used when calling a function by specifying the parameter name along with its value.

```
def kids(child3, child2, child1):
    print("The youngest is " + child3)

kids(child1="poly", child2="Tolly", child3="koly")
```

Output:

```
The youngest is koly
```

Arbitrary Keyword Arguments

Use **kwargs when you don’t know how many key=value pairs will be passed.

```
def student_details(**info):
    for key, value in info.items():
        print(key, ":", value)

student_details(name="Ayesha", age=16, grade="B",
city="Dhaka")
```

Output:

```
name : Ayesha
age : 16
grade : B
city : Dhaka
```

Default Parameter Value

```
def country(name="Norway"):
    print("I am from " + name)


country("Sweden")
country()
```

Output:

```
I am from Sweden
I am from Norway
```

Empty Function

```
def my_function():
    pass    # avoids error if no code inside
```

Return Values

```
def add(a, b):
    return a + b   # function sends back the sum


result = add(5, 3)   # store the returned value
print("Result is:", result)
```

- print() → only shows something on screen
- return → gives a value back so we can **reuse** it later

QUESTION SESSION

THANK YOU

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